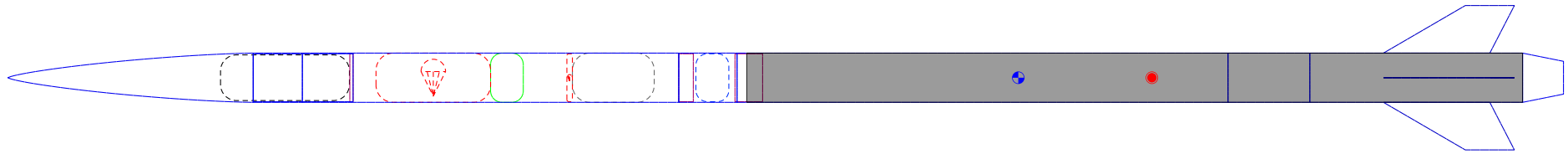


Rocket Design



Invictus 8.5

Stages: 1

Mass (with motor): 152 lb

Stability: 2.72 cal / 8.59 %

CG: 123 in

CP: 140 in

N27-30k(THRUSTMIKEY)-P

Altitude	31703 ft	Motor	Avg Thrust	Burn Time	Max Thrust	Total Impulse	Thrust to Wt	Motor Wt	Size
Flight Time	389 s	N27-	2784 N	14.3 s	3952 N	40134 Ns	4.11:1	47 lb	6/94.8 in
Time to Apogee	46.2 s	30k(THRUS							
Optimum Delay	29.4 s	TMIKEY)							
Velocity off Pad	101 ft/s								
Max Velocity	1504 ft/s								
Velocity at Deployment	92.3 ft/s								
Landing Velocity	19.6 ft/s								

Parts Detail

Stage: Sustainer



Nose Cone	Fiberglass (1.07 oz/in³)	Haack series	Len: 30 in	Mass: 1.91 lb
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Payload		Dia _{out} 5.568 in		Mass: 4.4 lb
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straight	Fiberglass (1.07 oz/in³)	Dia _{in} 5.843 in Dia _{out} 6 in	Len: 6 in	Mass: 0.587 lb
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Forward Body Tube	Carbon fiber (1.03 oz/in³)	Dia _{in} 5.843 in Dia _{out} 6 in	Len: 46 in	Mass: 4.33 lb
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Bulkhead pay	Carbon fiber (1.03 oz/in³)	Dia _{out} 5.843 in	Len: 0.4 in	Mass: 0.69 lb
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Main Chute	Ripstop nylon (0.22 oz/ft²)	Dia _{out} 144 in	Len: 14 in	Mass: 1.56 lb
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Shroud Lines	Elastic cord (round 2 mm, 1/16 in) (0.019 oz/ft)	Lines: 6	Len: 11.811 in	
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Recovery Bay		Dia _{out} 6 in		Mass: 6 lb
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Compact Elliptical Parachute [Cd 1.55 (3.3 oz) 18.0 in^3] Fruity Chutes CFC-036-S	Ripstop nylon, lightweight, 3 mil (0.117 oz/ft²)	Dia _{out} 36 in	Len: 0.673 in	Mass: 0.206 lb
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Shroud Lines Fruity Chutes CFC-036-S	Spectra #400 [Oval 2.55 mm, 3/32in] (0.013 oz/ft)	Lines: 12	Len: 41.4 in	
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Avionics Bay		Dia _{out} 6 in		Mass: 2 lb
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plumbing	Carbon fiber (1.03 oz/in³)	Dia _{in} 5.843 in Dia _{out} 6 in	Len: 7.12 in	Mass: 0.669 lb
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Bulkhead	Aluminum (1.56 oz/in³)	Dia _{out} 5.843 in	Len: 1.775 in	Mass: 4.64 lb
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top plumbing		Dia _{out} 5.843 in		Mass: 4 lb
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Bulkhead	Aluminum (1.56 oz/in³)	Dia _{out} 5.843 in	Len: 3.375 in	Mass: 0 lb
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Ox Tank (Undetermined)	Aluminum (1.56 oz/in³)	Dia _{in} 5.843 in Dia _{out} 6 in	Len: 60 in	Mass: 0 lb
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Plumbing Cover (Undertermined)

Carbon fiber
(1.03 oz/in³)

Dia_{in} 5.843 in
Dia_{out} 6 in

Len: 10 in

Mass: 0 lb



Combustion Chamber (Undetermined)

Aluminum
(1.56 oz/in³)

Dia_{in} 5.843 in
Dia_{out} 6 in

Len: 26 in

Mass: 0 lb



fins (4)

Carbon fiber
(1.03 oz/in³)

Thick: 0.3 in

Mass: 3.61 lb



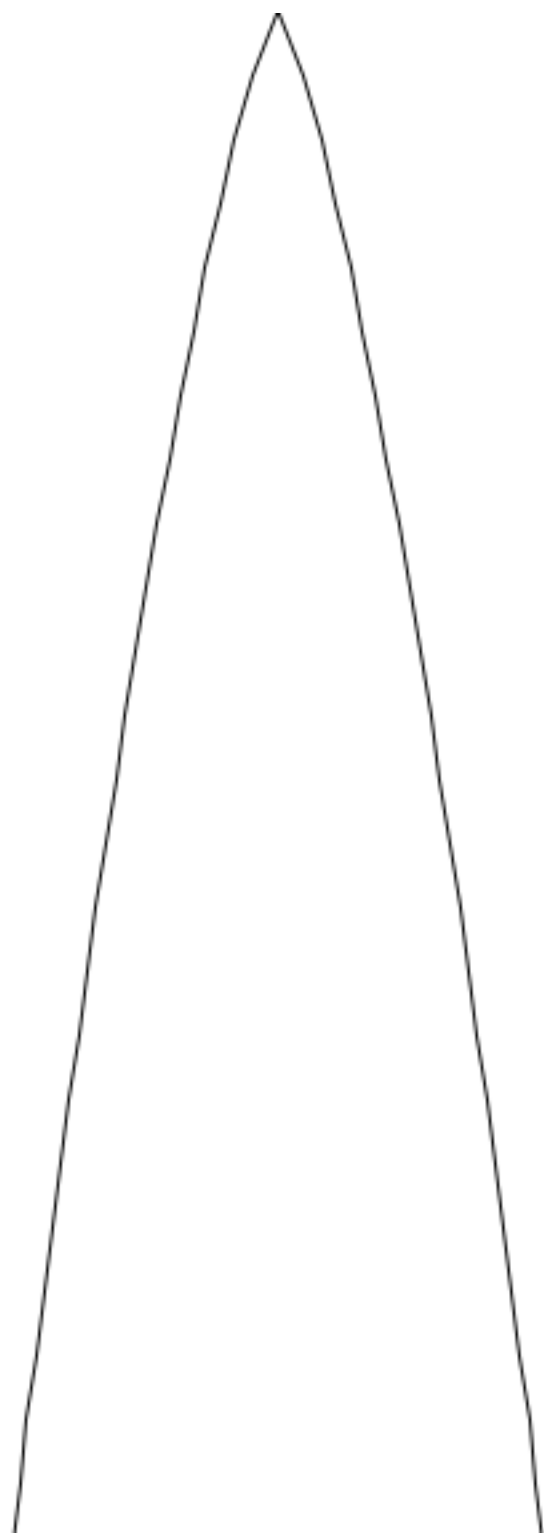
Transition

Aluminum
(1.56 oz/in³)

Fore Dia: 6 in
Aft Dia: 4 in

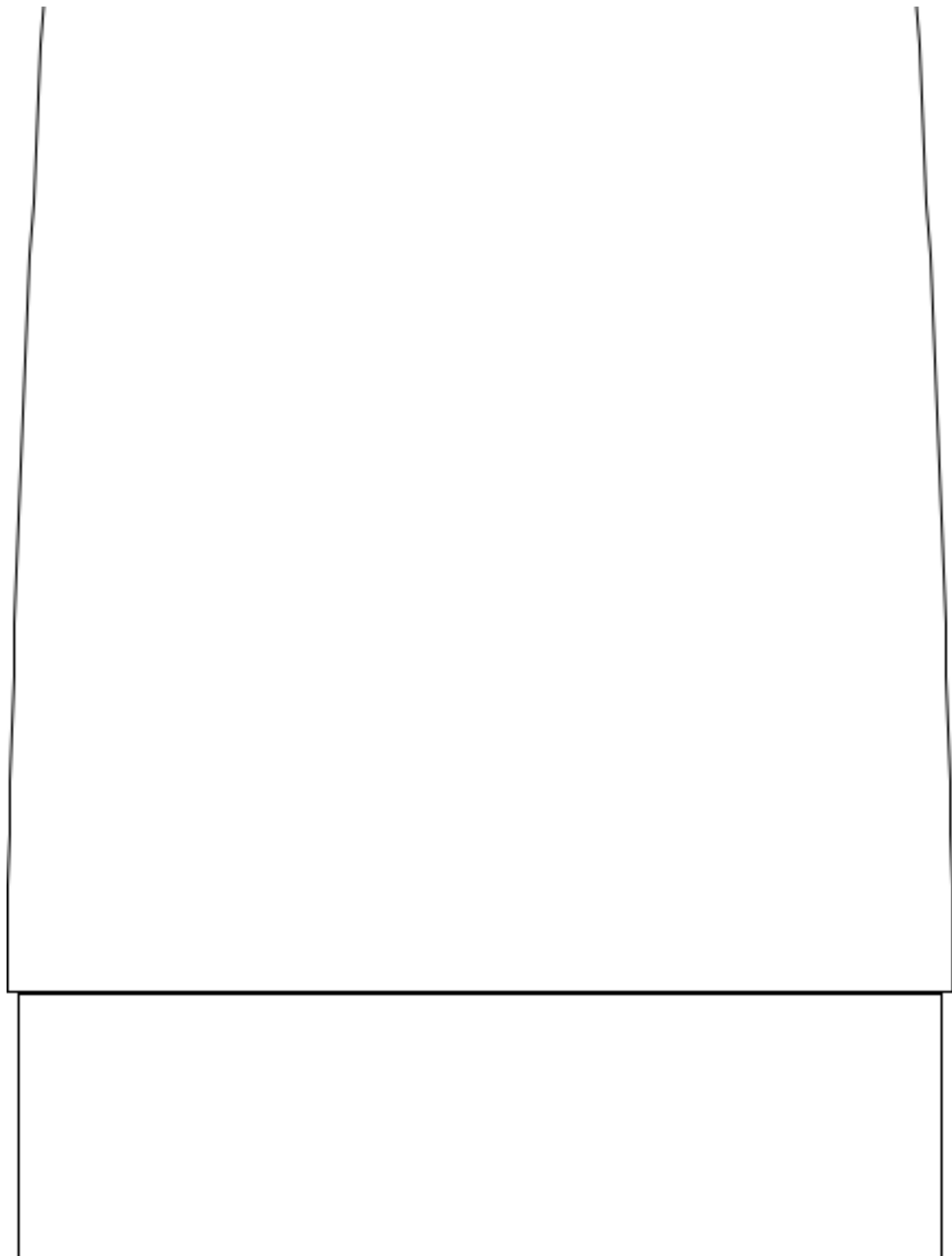
Len: 5 in

Mass: 1.5 lb





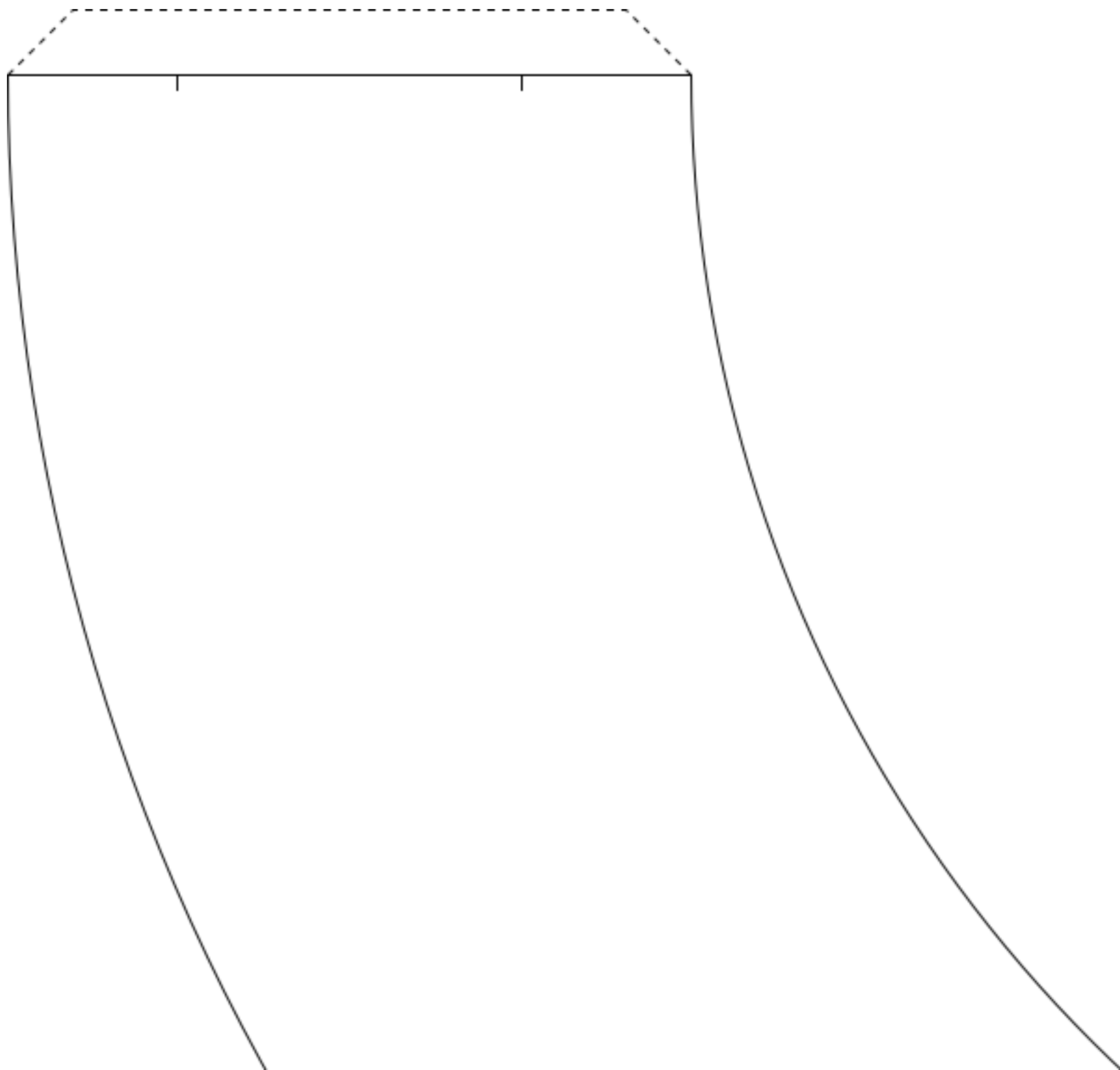


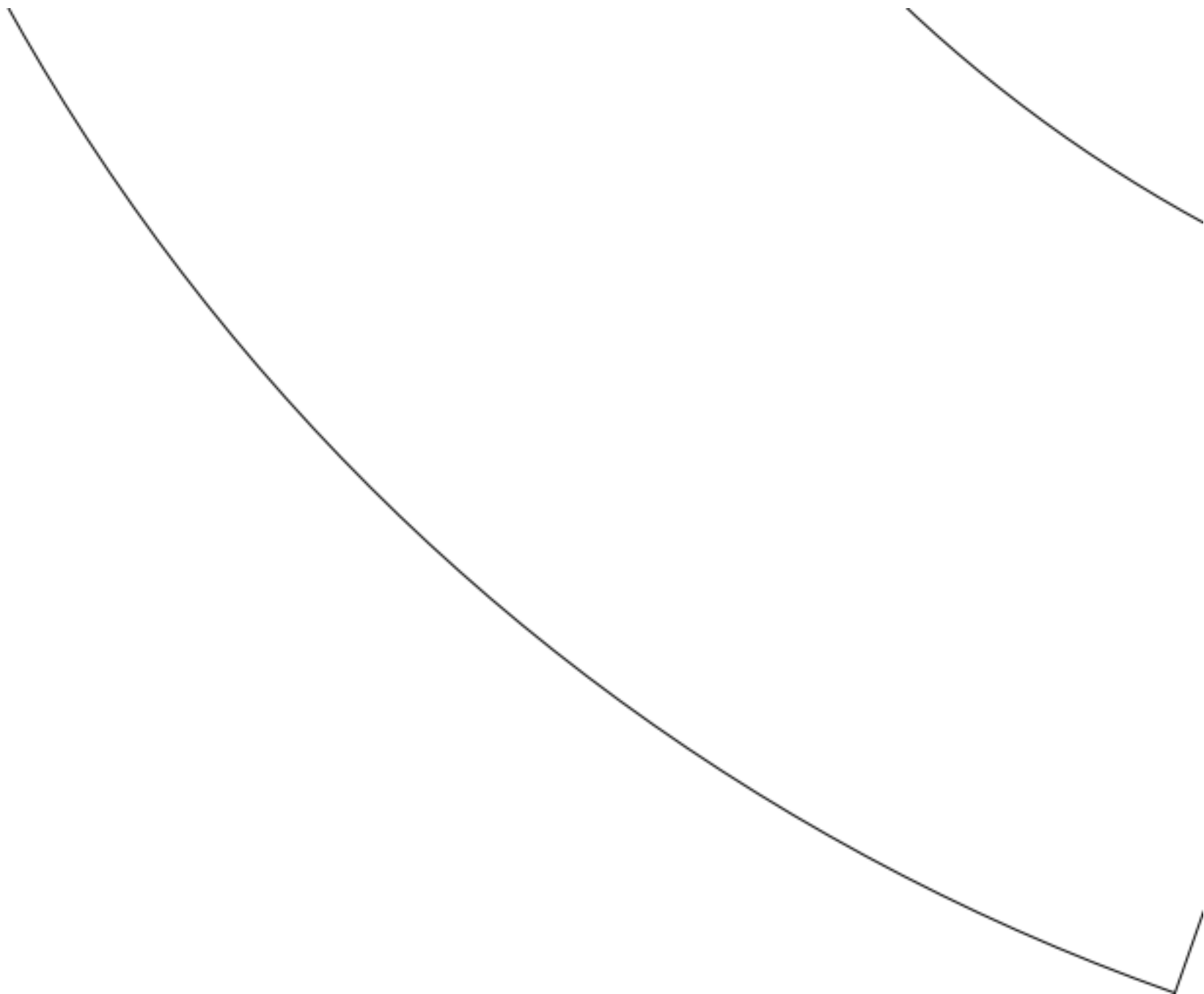


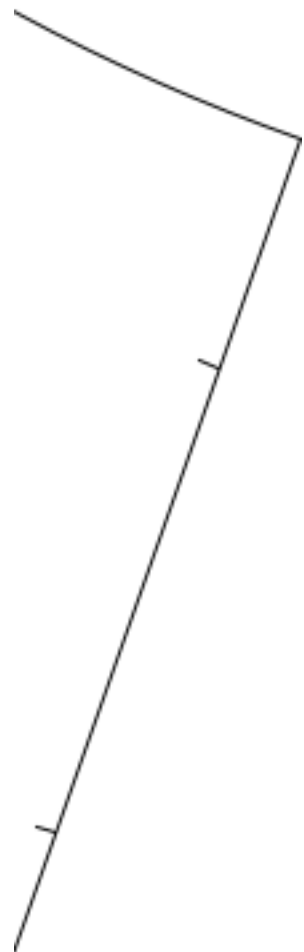
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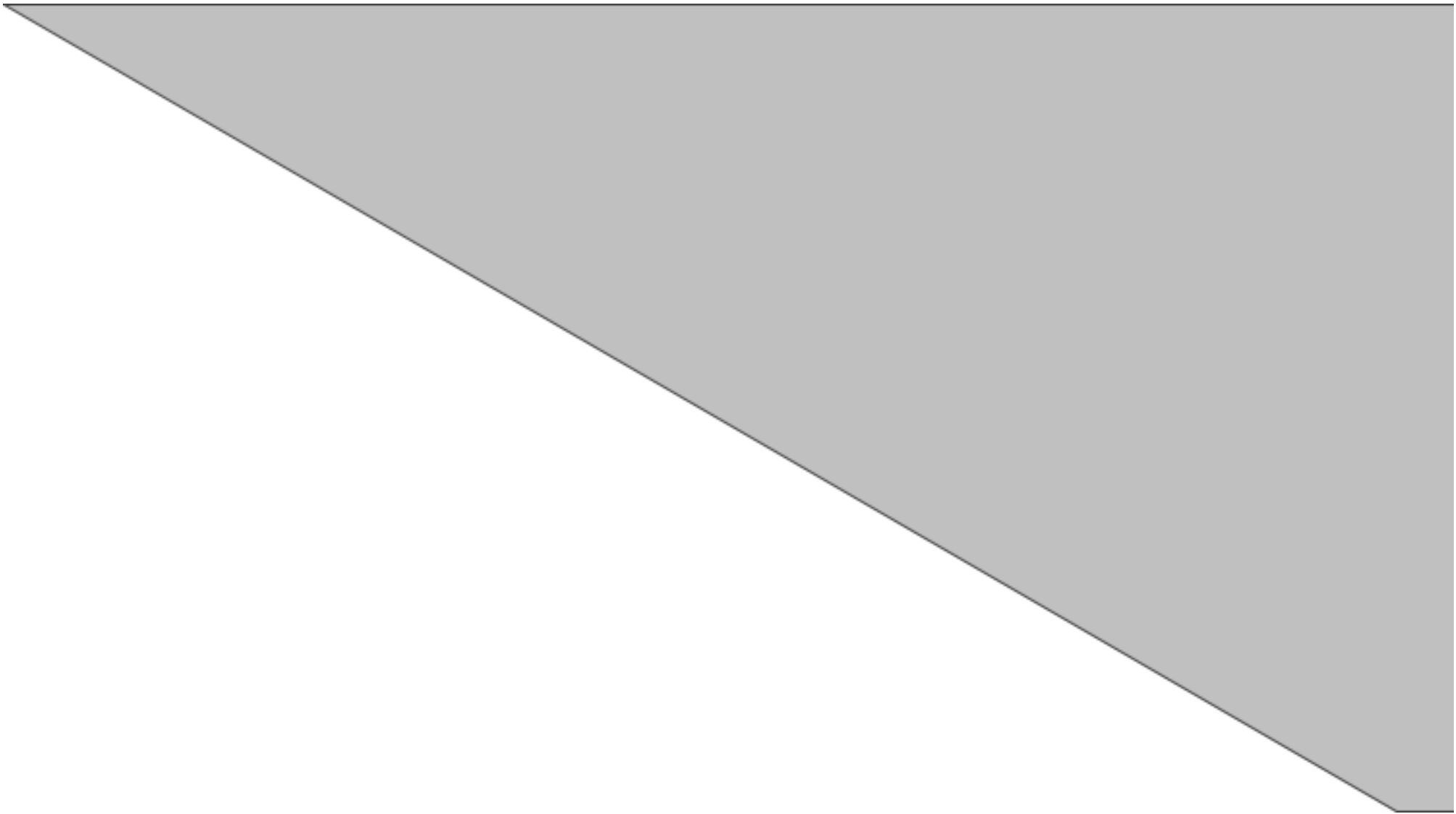
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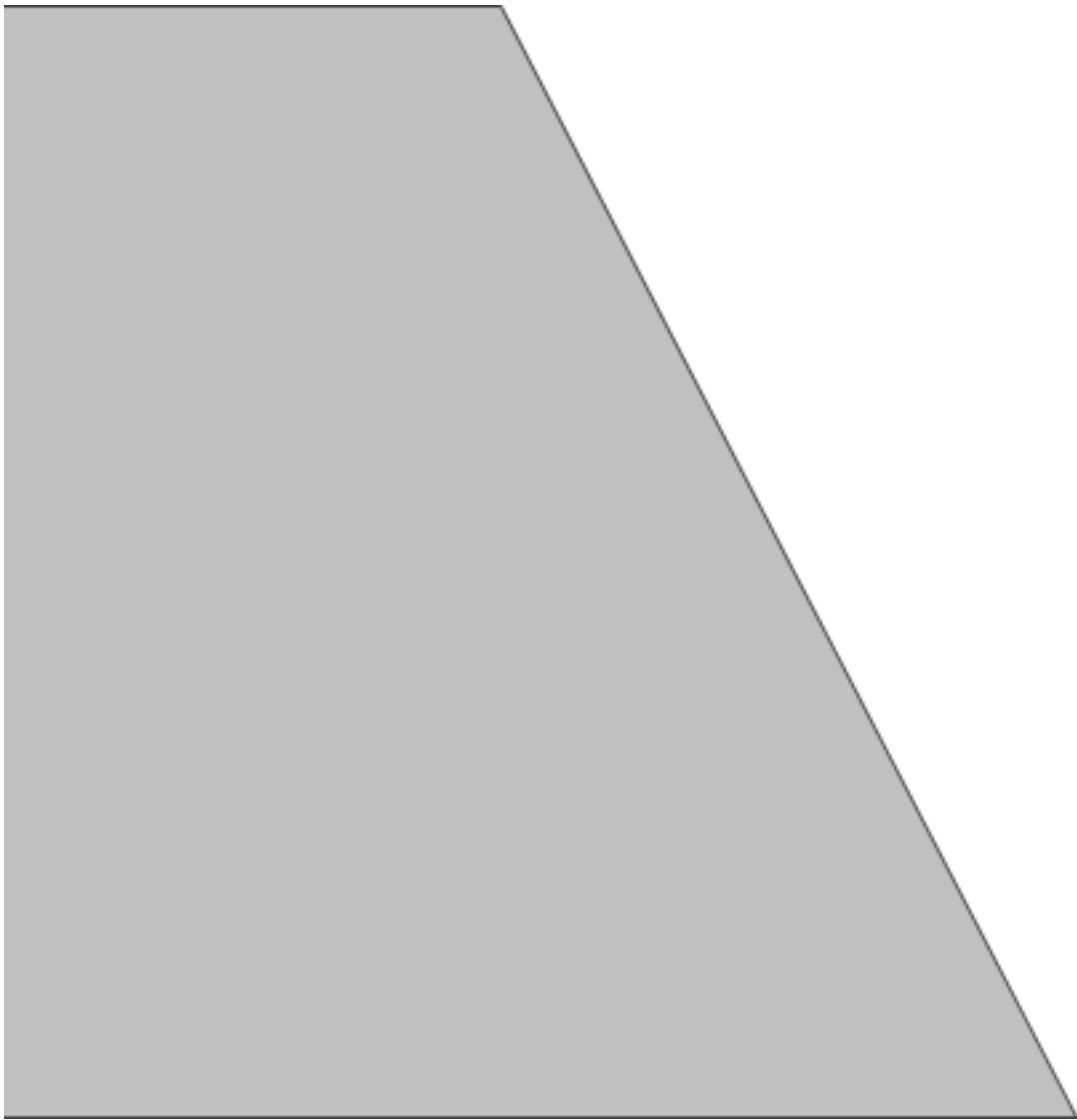


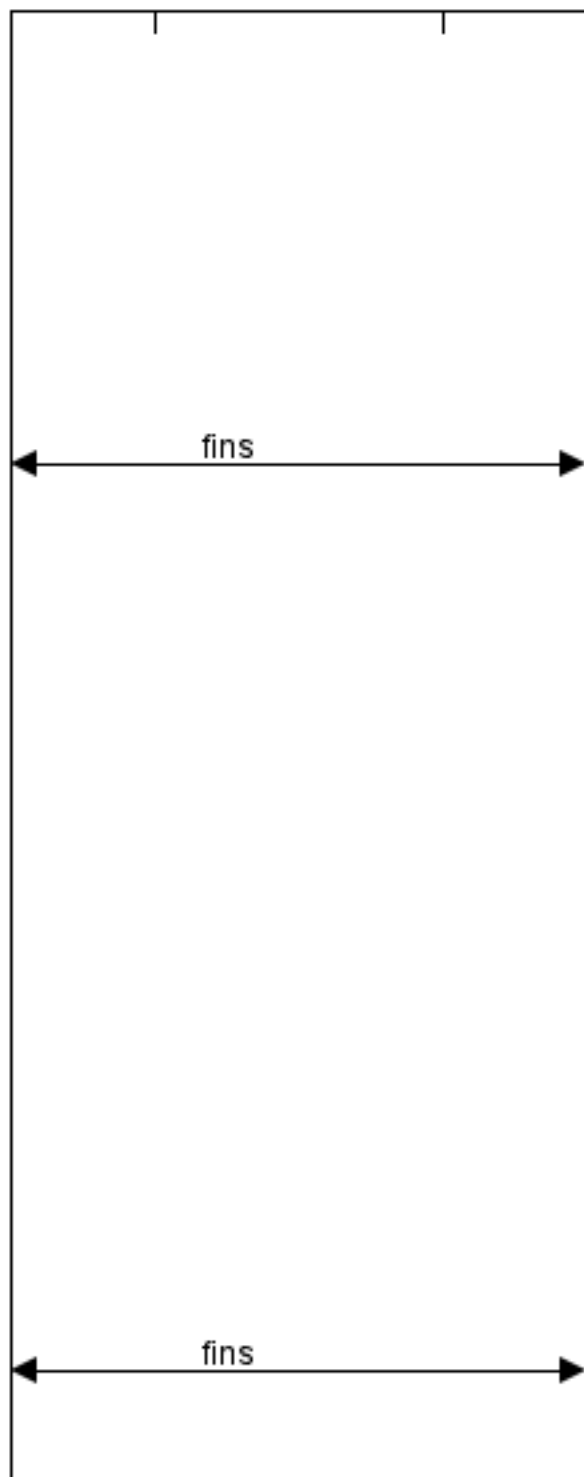


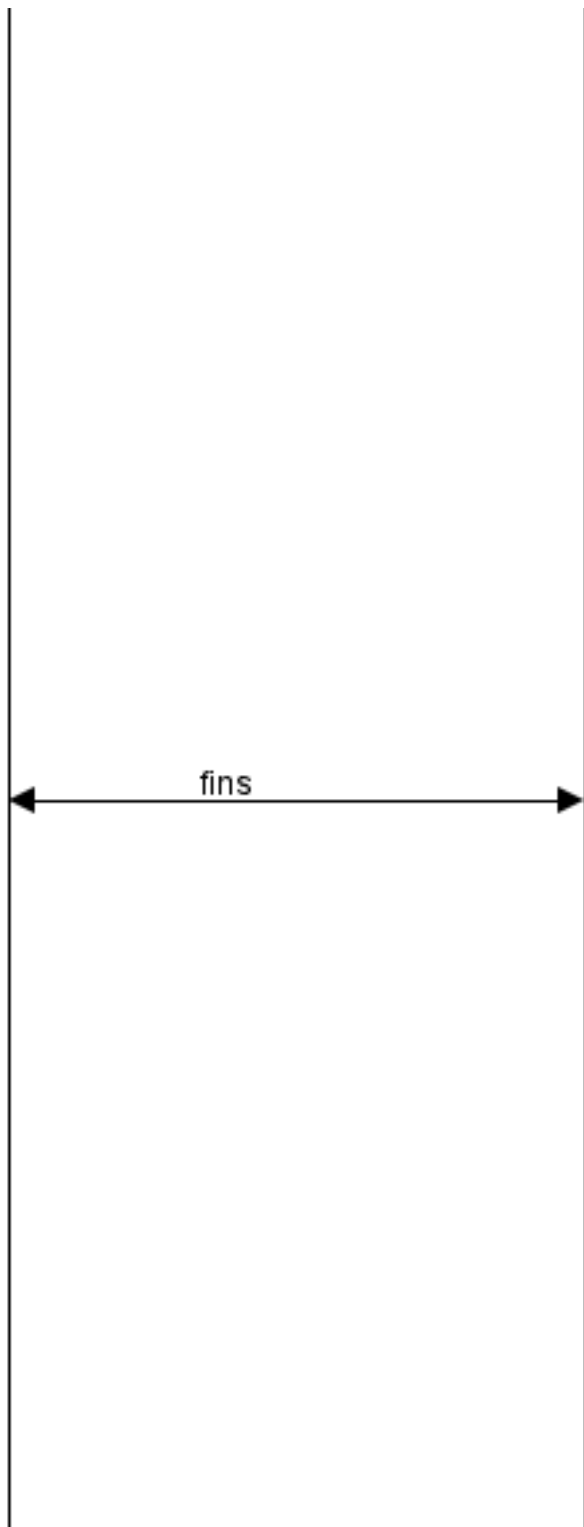


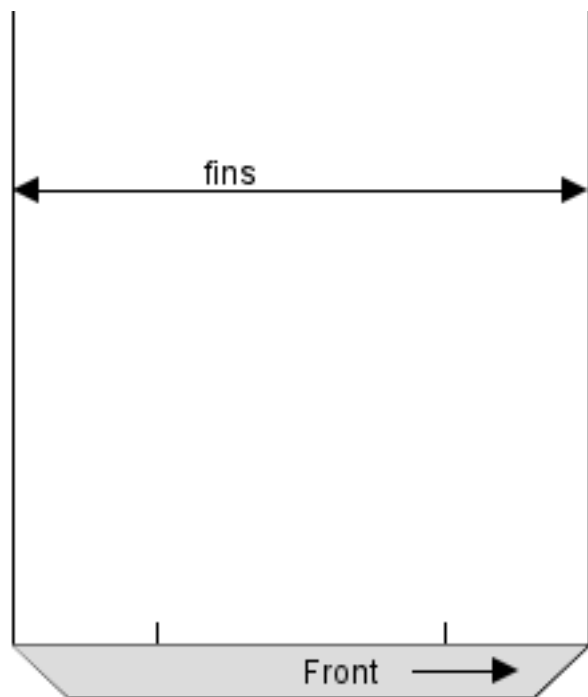












in cm

